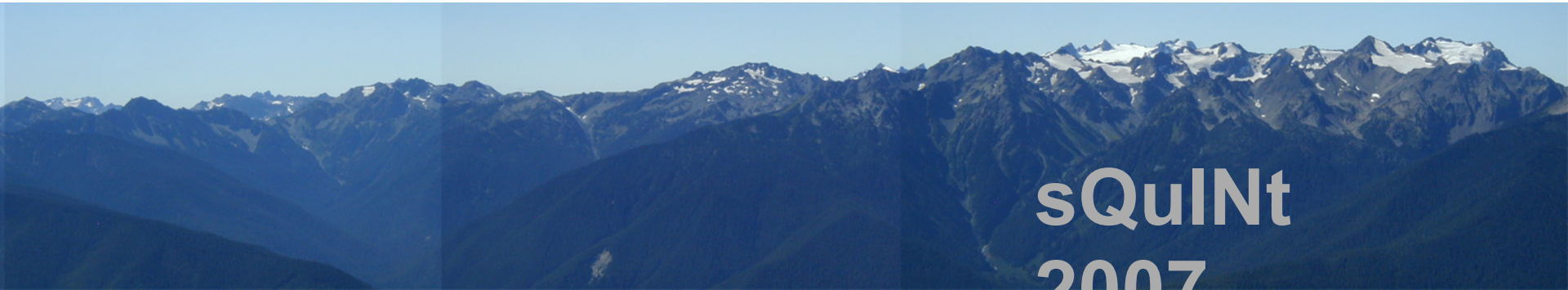
Is Fault-Tolerant Adiabatic Quantum Computation Possible?

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“P” Wave

Journalists Says: “Quantum computers, on the other hand, can evaluate all possible solutions simultaneously and find the optimal solution, often in just a few clock cycles, thereby not only vastly speeding up the time taken to find the solution but also finding the most optimal result.”

Question: Will we build a quantum computer before or after we run out of SQUINT spellings?

squint	sqUint	sQuint	sQUint	Squint	SqUint	SQuint	SQUint
squinT	sqUinT	sQuinT	sQUinT	SquinT	SqUinT	SQuinT	SQUinT
squiNt	sqUiNt	sQuiNt	sQUiNt	SquiNt	SqUiNt	SQuiNt	SQUiNt
squiNT	sqUiNT	sQuiNT	sQUiNT	SquiNT	SqUiNT	SQuiNT	SQUiNT
squInt	sqUInt	sQuInt	sQUInt	SquInt	SqUInt	SQuInt	SQUInt
squInT	sqUInT	sQuInT	sQUInT	SquInT	SqUInT	SQuInT	SQUInT
squINt	sqUINt	sQuINt	sQUINt	SquINt	SqUINt	SQuINt	SQUINt
squIN T	sqUINT	sQuINT	sQUINT	SquIN T	SqUINT	SQuINT	SQUINT

Ponds

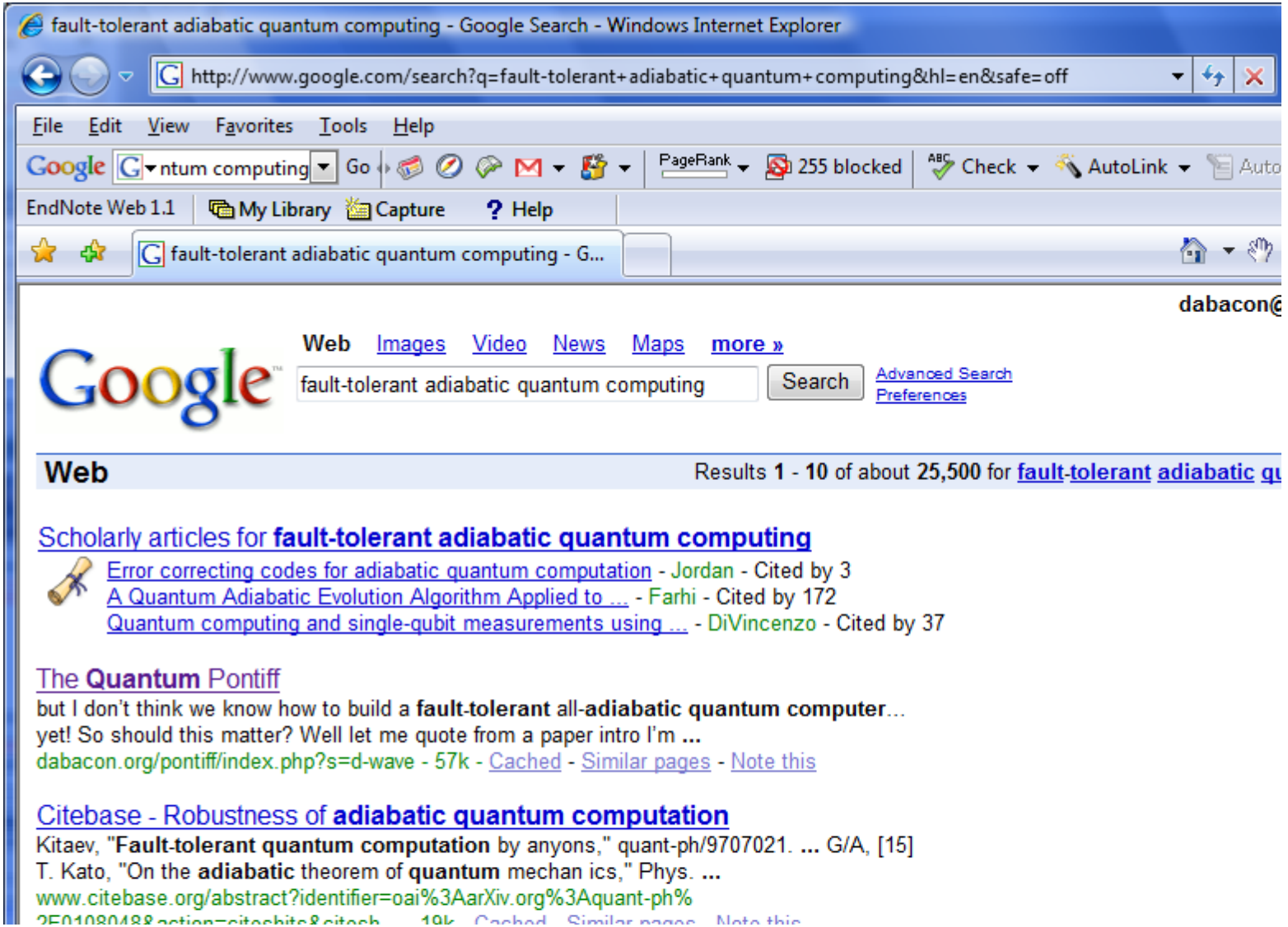








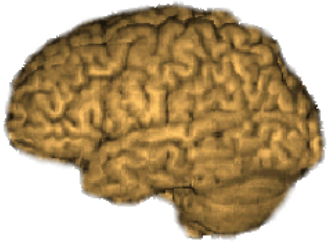
Is Fault-Tolerant Adiabatic Quantum Computation Possible?



Two Views of Information

Information is "fungible" adj. being of such a nature that one part or quantity may be replaced by another equal part or quantity in the satisfaction of an obligation.







Al Gore

George Bush

Ralph Nader

Pat Buchanan

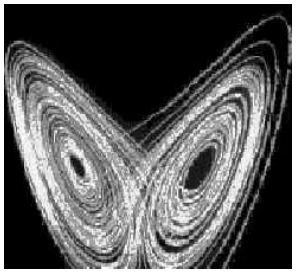




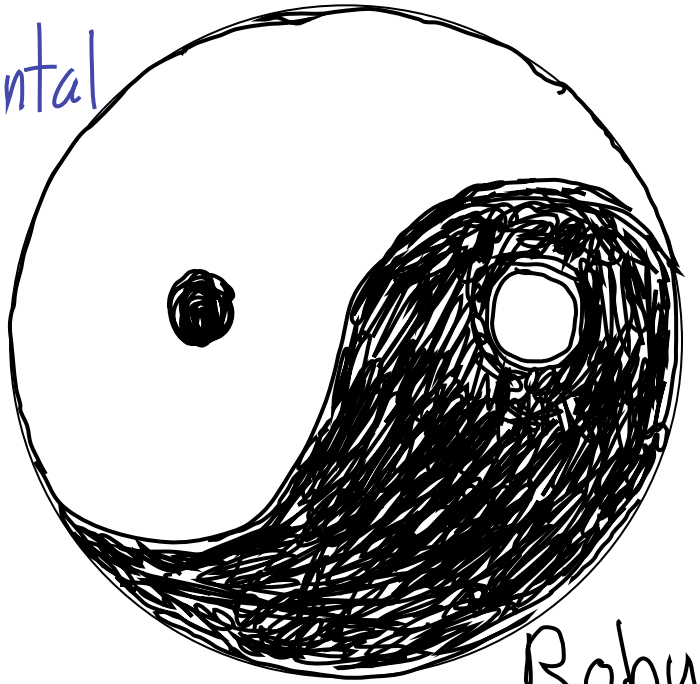
Rolf Landauer

No Information without Physical Representation!

Not all physical systems are robust information processing machines



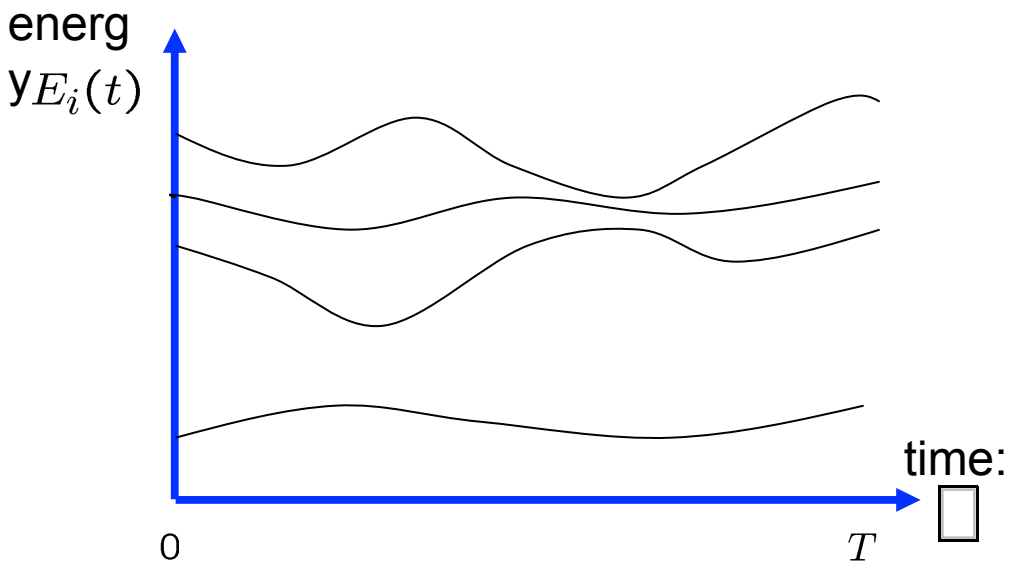
Tao of Information



This is a Jealous Slide

A Beautiful Story

Time dependent Hamiltonian:



$$t = sT$$

$$\mathcal{F} = \max_{m \neq n} \max_{0 \leq s \leq 1} |\langle \psi_m(s) | \frac{dH}{ds} | \psi_n(s) \rangle|$$

$$\mathcal{G} = \min_{m \neq n} \min_{0 \leq s \leq 1} |E_m(s) - E_n(s)|$$

Adiabatic condition:

$$T \gg \frac{\mathcal{F}}{\mathcal{G}^2}$$

Adiabatic Quantum Algorithms

[E. Farhi, J. Goldstone, S. Gutmann, and M. Sipser, 2000]

$$H(s) = (1-s)H_0 + sH_p \qquad 0 \le s \le 1$$

1. Prepare ground state of H_0
2. "Slowly" evolve $H(s)$
3. Measure final state, obtaining answer to problem instance *p*

Example: $H_0 = J_z \sum_i (\sigma_x)_i \quad |\psi_0(0)\rangle = \left[\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right]^{\otimes n}$

$$H_p = \sum_i J_i(p)(\sigma_z)_i + \sum_{<i,j>} J_{i,j}(p)(\sigma_z)_i(\sigma_z)_j$$

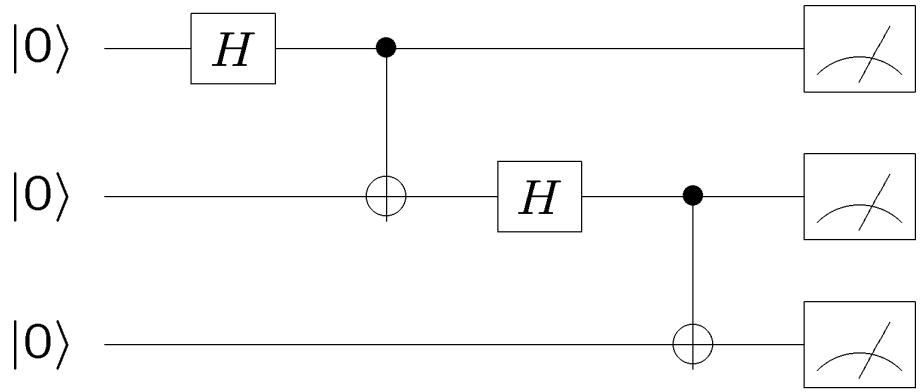
Finding ground state of 2D Ising in magnetic field when bond values are problem instance is NP-complete [Barahona 82]

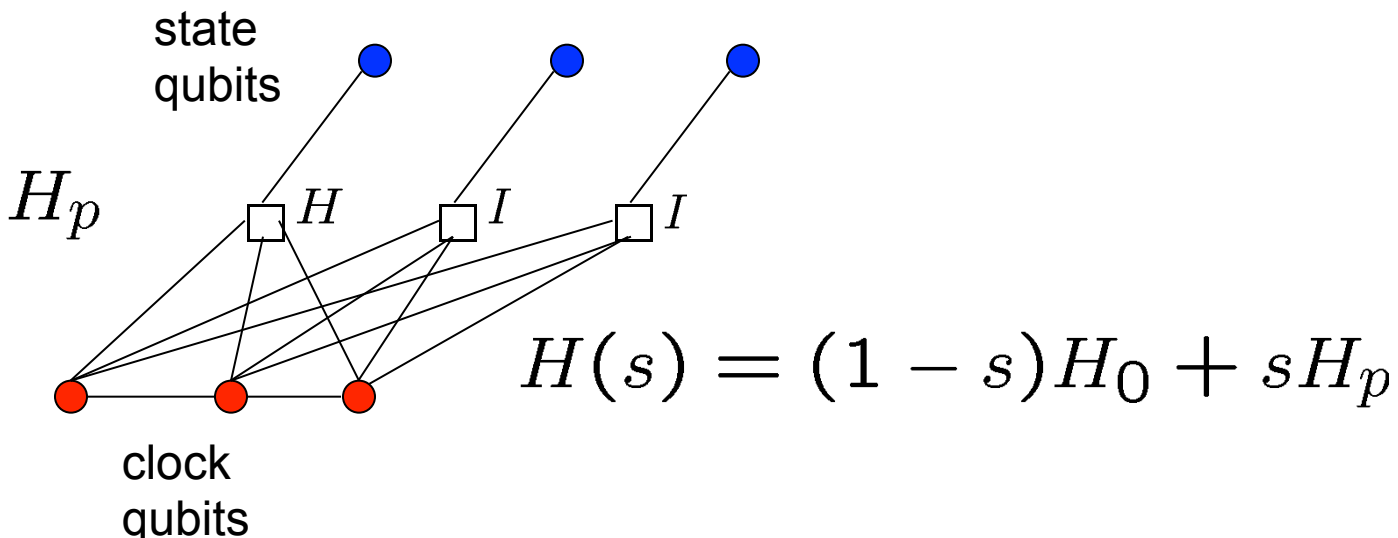
Power of adiabatic algorithms? **Much Debate!**

Universal Adiabatic Quantum Computation

Adiabatic quantum computation ⇔ The quantum circuit model

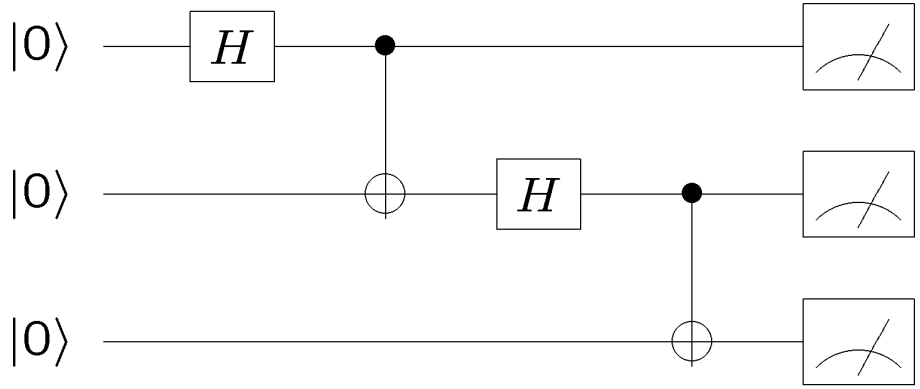
[D. Aharonov, W. van Dam, J. Kempe, Z. Landau, S. Lloyd, O. Regev, 2004]

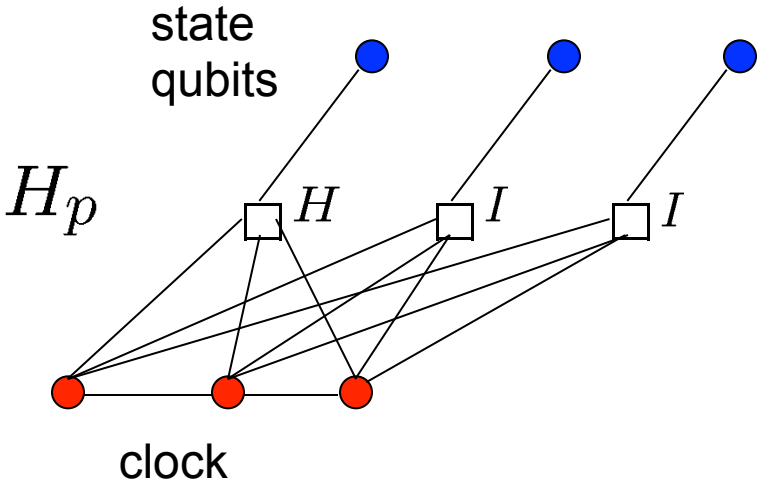




$$H(s) = (1-s)H_0 + sH_p$$

Universal Adiabatic Quantum Computation





$$H(s) = (1-s)H_0 + sH_p$$

Final ground state is history of circuit:

$$|\psi_0\rangle = \frac{1}{\sqrt{s+1}} \sum_{i=0}^{s} |\psi(i)\rangle \otimes |i\rangle_{clock} \qquad |\psi(i)\rangle = U_i U_{i-1} \cdots U_1 |\psi(0)\rangle$$

Pad circuit with identities to boost amplitude of answer

Gap is 1/poly(circuit size)

Towards an Architecture?

3-qubit Hamiltonians, 2-qu6it Hamiltonians on a 2D lattice

[D. Aharonov, W. van Dam, J. Kempe, Z. Landau, S. Lloyd, O. Regev, 2004]

2-qubit Hamiltonians

[J. Kempe, A. Kitaev, O. Regev, 2004]

2-qubit Hamiltonians in a plane

[R. Oliveira, B. Terhal 2005]

2-body Hamiltonians in a plane with spatial degrees of freedom

[A. Mizel, D.A. Lidar, M. Mitchell 2006]

A new **architecture** for building a quantum computer?!?!



No Information without Physical Representation!

Rolf Landauer

Count the Ways

1. Preparation may be faulty.

2. Open quantum system, so adiabatic theorem still holds?

3. Open quantum system, so decoherence/noise?

4. Open quantum system, do we have control?

5. Faulty measurement

Is Fault-Tolerant Adiabatic Quantum Computation Possible?

Count the Ways

2. Open quantum system, so adiabatic theorem still holds?

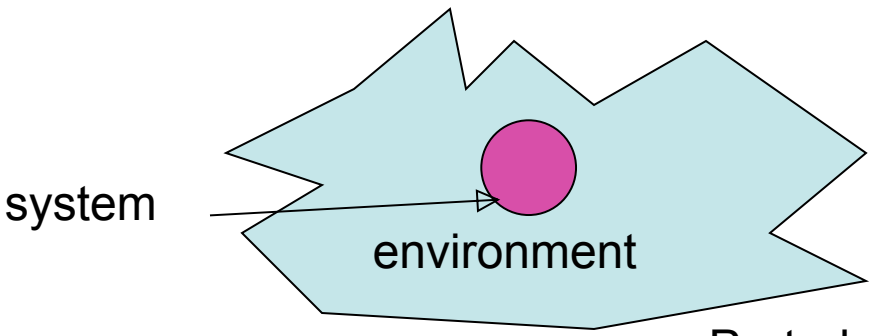
M. S. Sarandy and D. A. Lidar 2005
P. Thunstrom, J. Åberg, and E. Sjöqvist 2005

Open system adiabatic theorems.

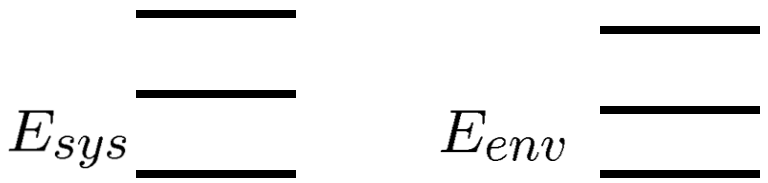
The longer you take, the more likely the effects of the broadening of your energy levels will shrink the gap and destroy the adiabatic condition.

"Static" Limit reminder: duck shoe thrown by Daniel Lidar

Energetics of Decoherence

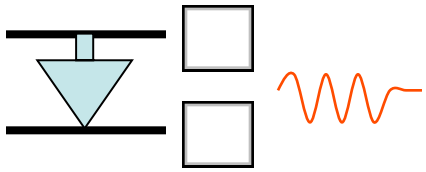


$$H = H_{sys} + H_{int} + H_{env}$$



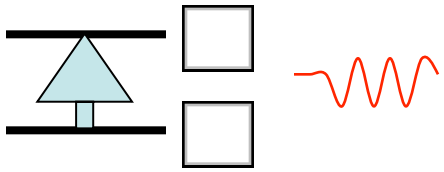
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy (rotating wave approx)



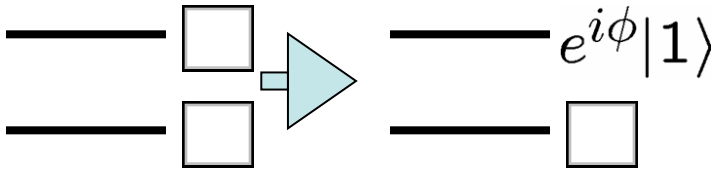
cooling

$\Delta E_{sys} < 0$



heating

$\Delta E_{sys} > 0$



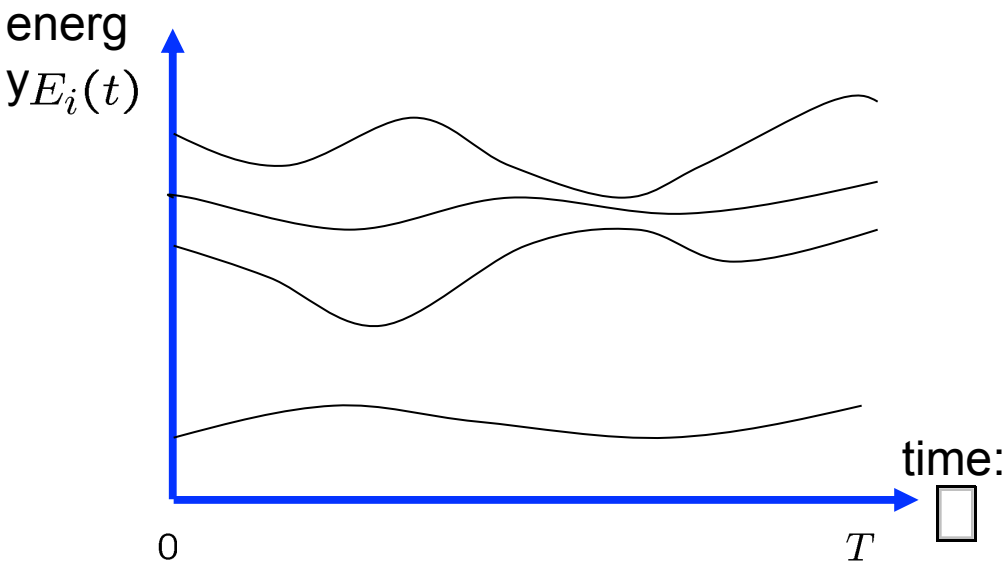
dephasing

$\Delta E_{sys} = 0$

by **cooling environment** we can eliminate heating

An Early Result

[Childs, Farhi, Preskill 2001]



If bath temperature is much less than energy gap: safe

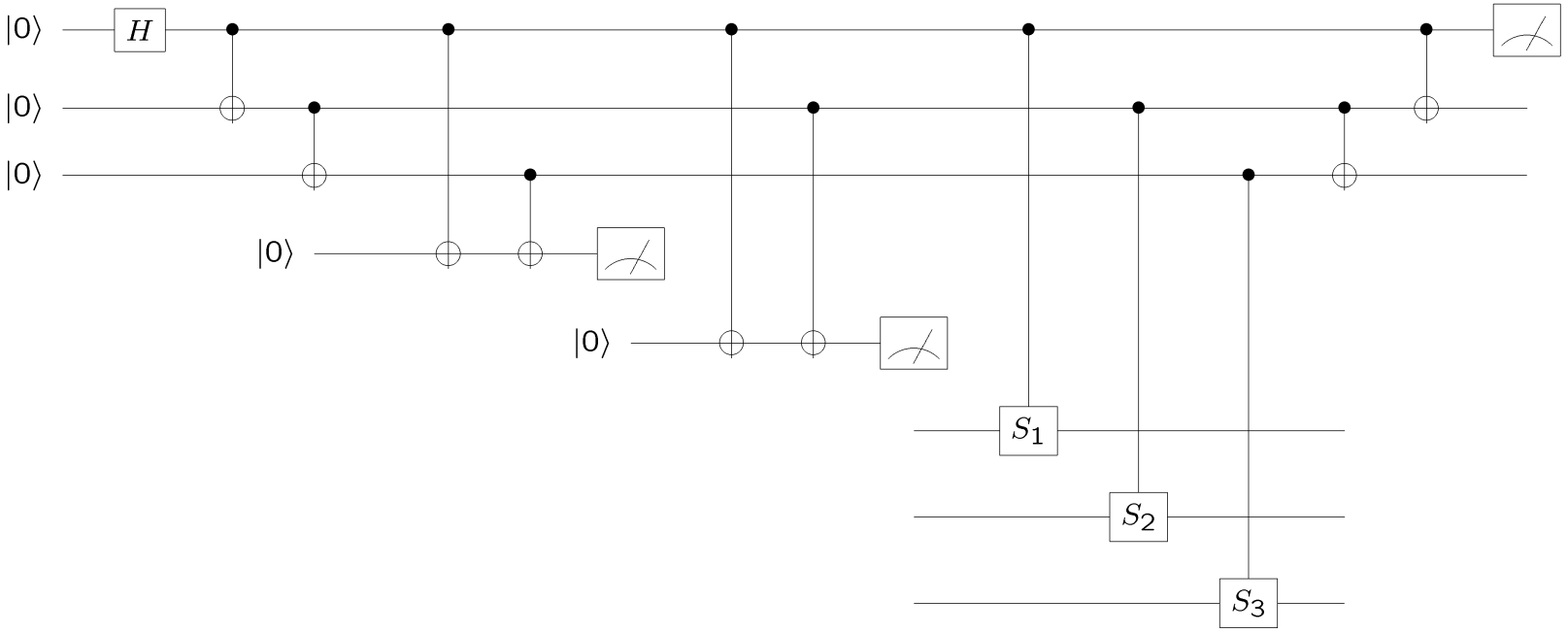
Problem: Temperature will have to shrink like 1/poly(n)



Port?



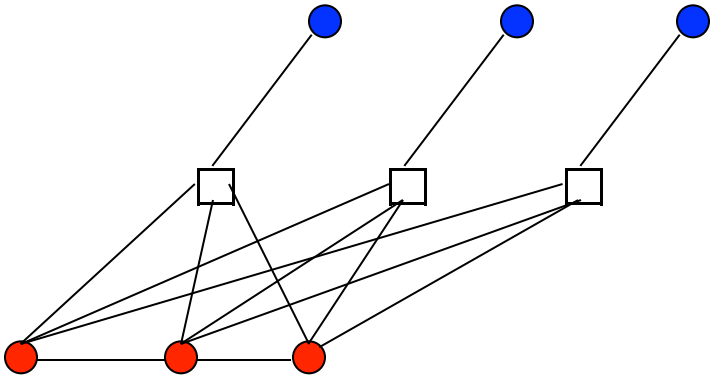
Standard QEC -> Hamiltonian Form

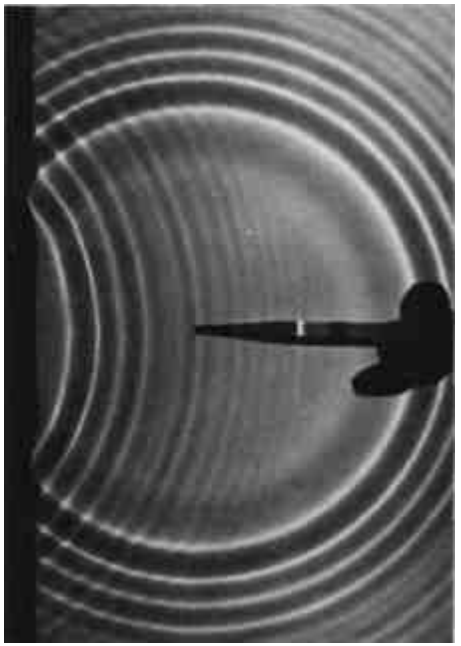


This can care for certain types of error.

Example: Hamiltonian implements wrong unitary gate

The Puddle





Problem: local error can propagate through network

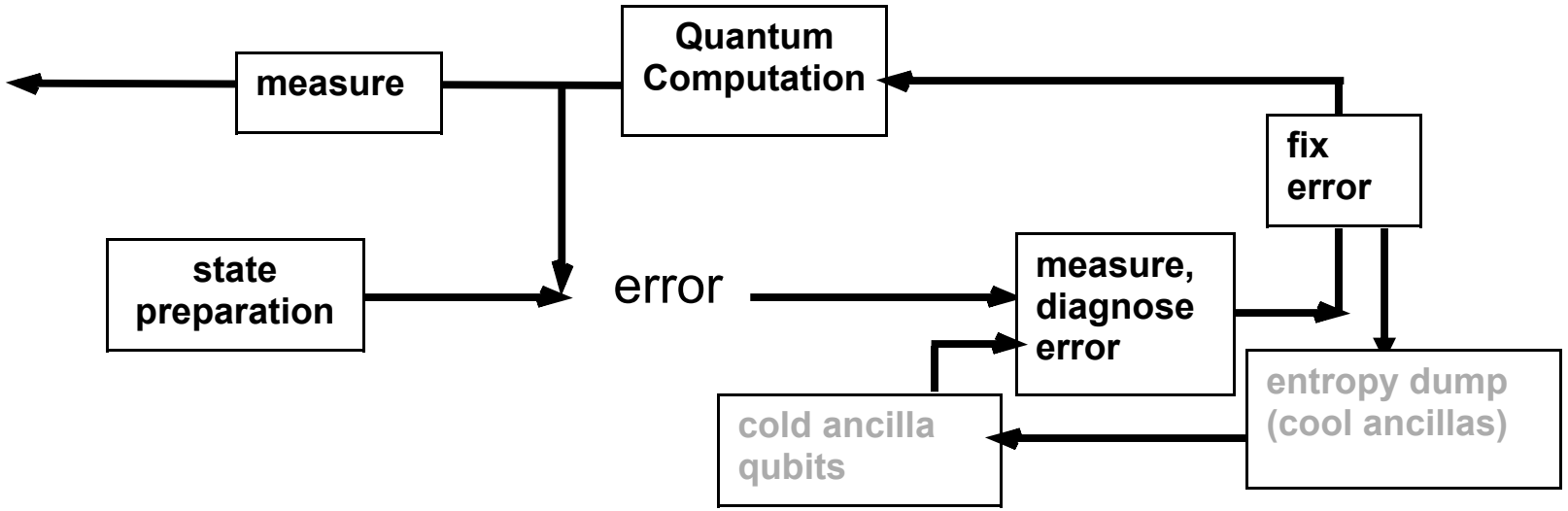
$$|\psi_0\rangle = \frac{1}{\sqrt{s+1}} \sum_{i=0}^{s} |\psi(i)\rangle \otimes |i\rangle_{clock}$$

Lessons of Quantum Error Correction

1. Information is in the eye of the beholder

$$s_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$s_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$s_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$s_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

2. Quantum error



A Later Result

[Jordan, Farhi, Shor 2006]

Stabilizer generators [[4,1,2]] code

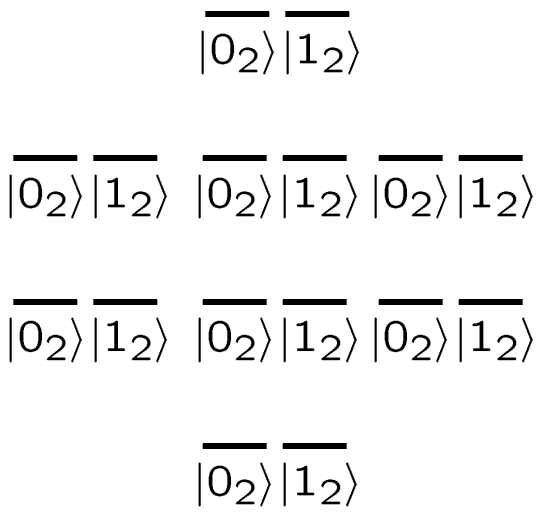
$$g_1 = X \otimes X \otimes X \otimes X$$
$$g_2 = Z \otimes Z \otimes Z \otimes Z$$
$$g_3 = X \otimes Y \otimes Z \otimes I$$

$$X_L = I \otimes X \otimes X \otimes I$$
$$Y_L = Y \otimes I \otimes Y \otimes I$$
$$Z_L = Z \otimes Z \otimes I \otimes I$$

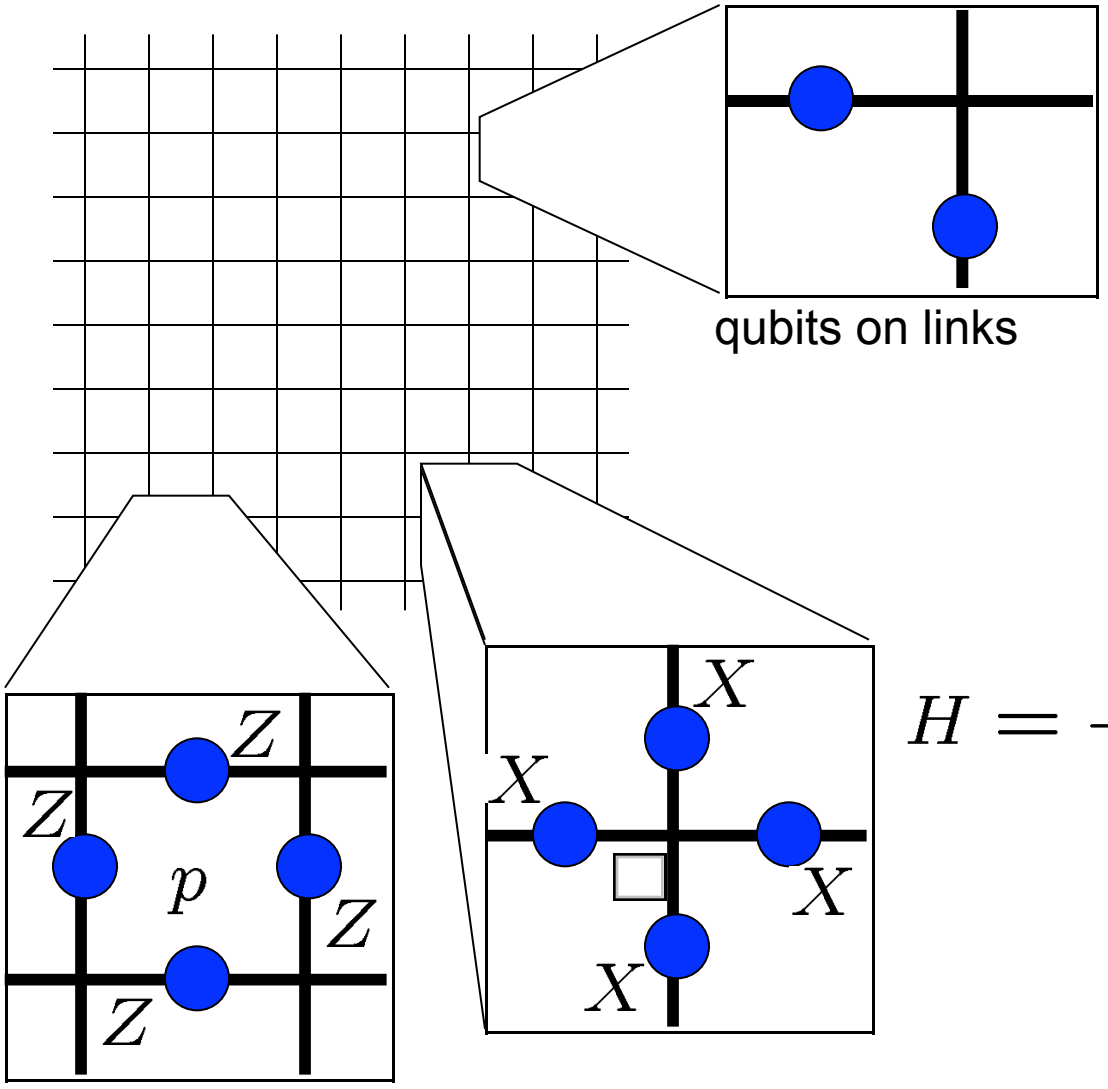
Hamiltonian:

$$H = -\frac{\Delta}{2}(g_1 + g_2 + g_4)$$

All single qubit errors
(X,Y,Z on single qubit)
take ground state to
excited state



Kitaev's Model



plaque operators:

$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$

$$H = -\frac{\Delta}{2}\left(\sum_p S_p + \sum_v S_v\right)$$

Problemo #1

Jordan, Farhi, Shor 2006

Protect against 1 qubit errors, 3 qubit interactions
To make universal for AQC, 4 qubit interactions

Protect against 2 qubit errors, 4 qubt interactions
To make universal for AQC, 6 qubit interactions

Kitaev's Toric Codes

Protect against L qubit errors, 4 qubit interactions
To make universal for AQC, 2L qubit interactions



Problemo #2

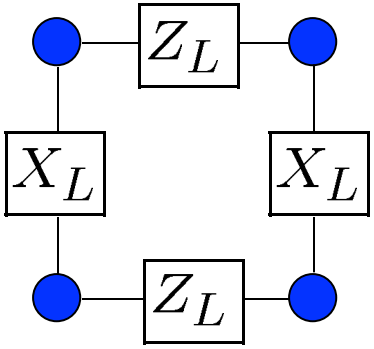
Geometry:

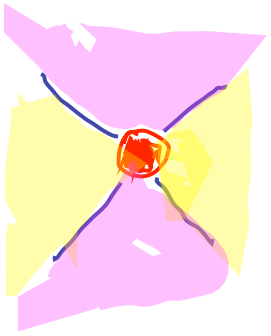
Stabilizer generators [[4,1,2]] code

$$g_1 = X \otimes X \otimes X \otimes X$$
$$g_2 = Z \otimes Z \otimes Z \otimes Z$$
$$g_3 = X \otimes Y \otimes Z \otimes I$$

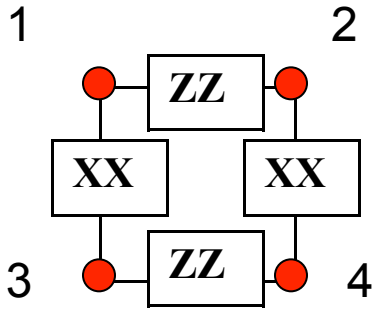
$$X_L = I \otimes X \otimes X \otimes I$$
$$Y_L = Y \otimes I \otimes Y \otimes I$$
$$Z_L = Z \otimes Z \otimes I \otimes I$$







Two Qubit Interactions?



Hamiltonian

$$H = -(Z_1 Z_2 + Z_3 Z_4 + X_1 X_3 + X_2 X_4)$$

Quantum error detecting code

$$S_1 = Z_1 Z_2 Z_3 Z_4$$

$$S_2 = X_1 X_2 X_3 X_4$$

$$S_i |\psi_C\rangle = |\psi_C\rangle$$

$$\bar{Z}_1 = Z_1 Z_2 \qquad \bar{X}_1 = Z_1 Z_3$$

$$\bar{Z}_2 = Z_1 Z_3 \qquad \bar{X}_2 = X_1 X_2$$

Code is a symmetry of the Hamiltonian:

$$[H, S_1] = [H, S_2] = 0$$

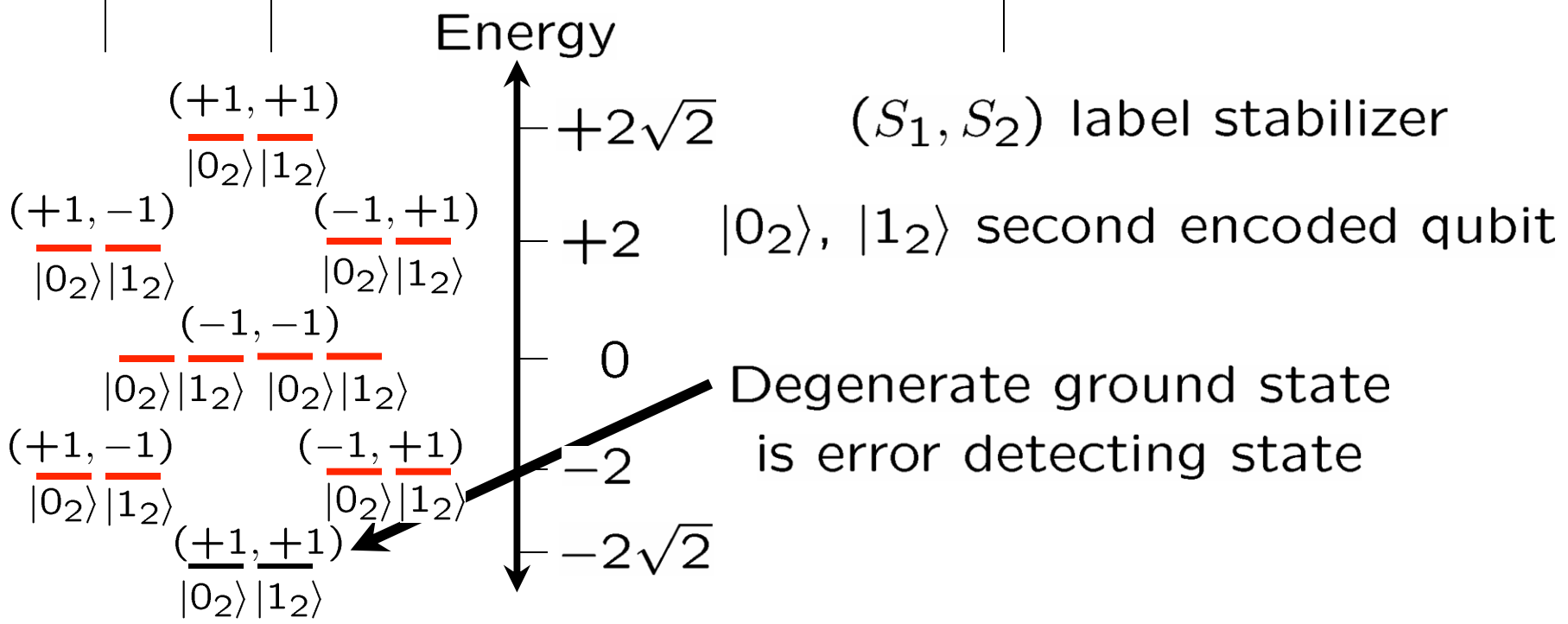
Code detects any single qubit error:

$$\{P, S_i\} = 0 \qquad P \text{ e.g. } Q \otimes I \otimes I \otimes I$$

H does not depend on second qubit (use to store encoded qubit)

$$H = -\bar{Z}_1 (I + S_1) - \bar{X}_1 (I + S_2)$$

S_1	S_2	$H = \bar{X}_1(1+S_1) + \bar{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\bar{X}_1 + \bar{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\bar{X}_1$	± 2
-1	$+1$	$2\bar{Z}_1$	± 2
-1	-1	0	0

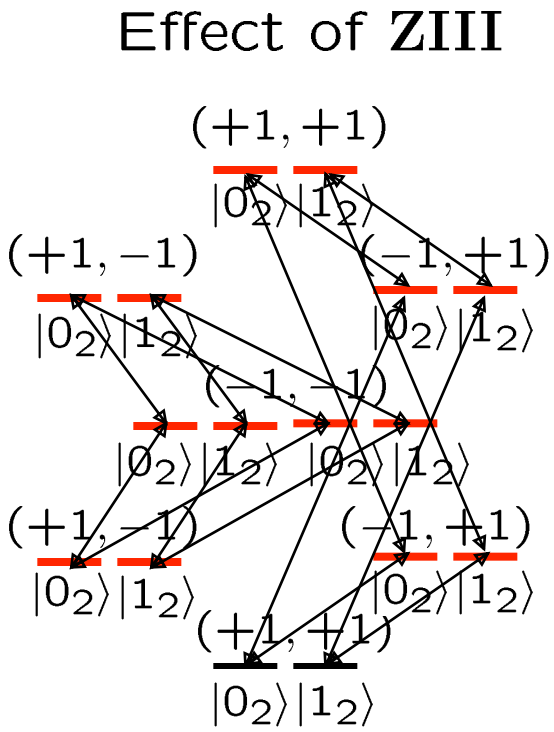


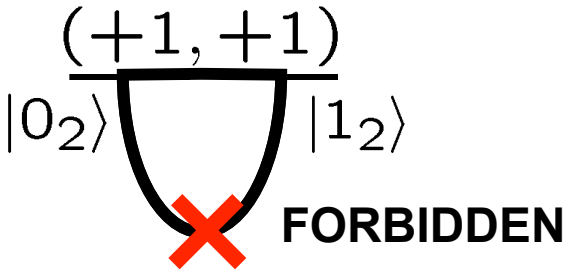
ground state is +1 eigenvalue of four qubit interactions

Supercoherence

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.





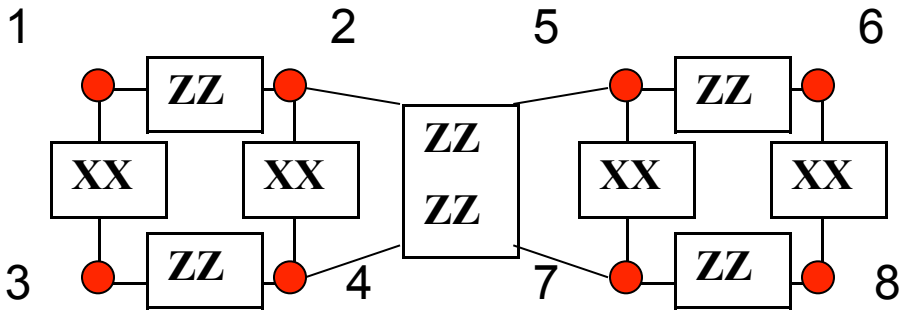
To first order all single qubit decoherence becomes heating

Energy gap and low temperature suppress lowest order decoherence

[Bacon, Brown, Whaley 2001]

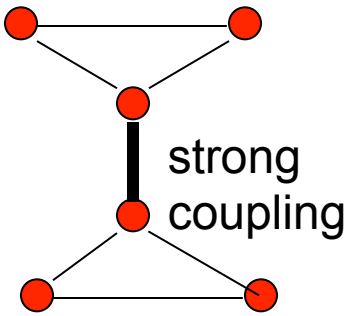
Two Qubit Encoded Terms?

$Z_A Z_B$



Uses 4 qubit interaction.

Can construct using “perturbation gadget”?



XZ – Hamiltonians?

Controlled-NOT plus real symmetric G such that G2 does not preserve the computational basis [Shi 03]

$$C_X = (I + Z) \otimes I + (I - Z) \otimes X$$

$$G = \cos tX + \sin tZ$$

Hamiltonian in universality proof [Aharonov et al 04] when using this gate set can be expanded in basis with only X and Z Pauli operators

Gadgets necessary to get this down to two qubit interactions can also be expanded in basis with only X and Z Pauli operators [Kempe 04]

XZ-Hamiltonians are Universal for AQC

Geometry

Can we do this in 2d plane?

Clock qubits: $|0_t\rangle = |000000\cdots0\rangle$

$|1_t\rangle = |100000\cdots0\rangle$

$|2_t\rangle = |110000\cdots0\rangle$

$|3_t\rangle = |111000\cdots0\rangle$

Ground state of $H = \Delta \sum_{i=0}^{n-1} |01\rangle_{i,i+1}\langle 01|_{i,i+1}$

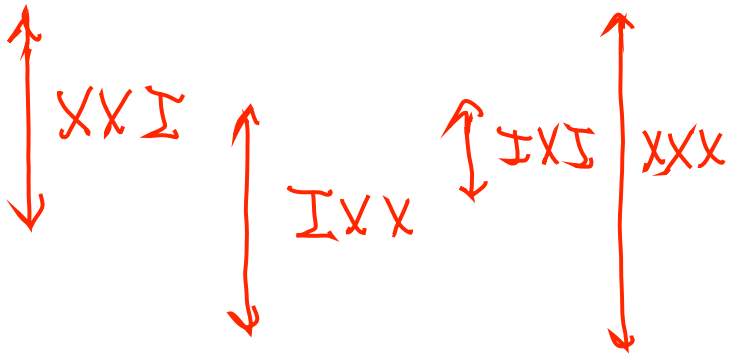
Use clock qubits as spatial degree of freedom:

$|0\rangle_0 = |000000\cdots0\rangle$

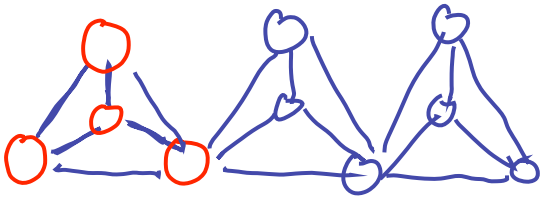
$|1\rangle_0 = |110000\cdots0\rangle$

$|0\rangle_1 = |111000\cdots0\rangle$

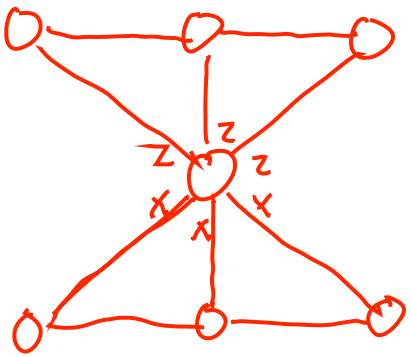
$|1\rangle_1 = |111100\cdots0\rangle$

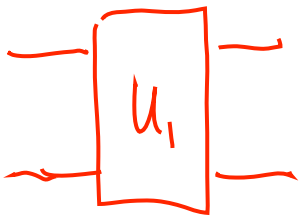


Geometry



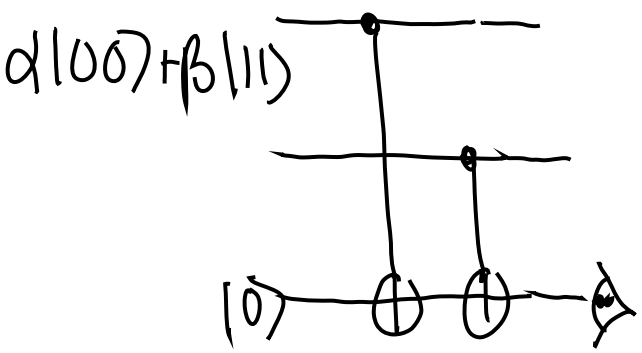




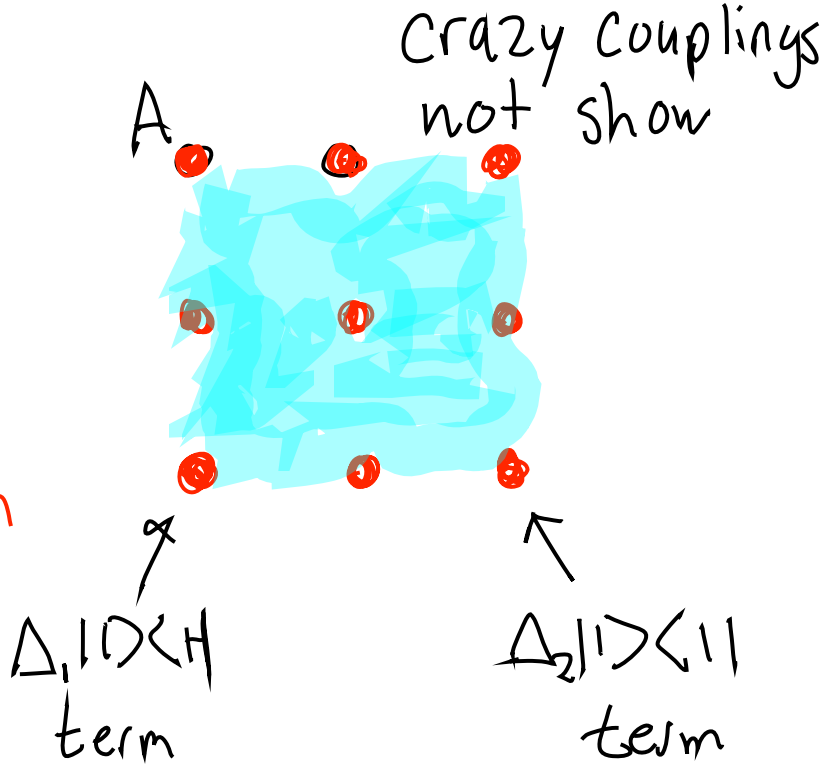


But to use any of the known QEC ideas, geometry becomes (at least) 3D

A Final Word



convert to AQC form

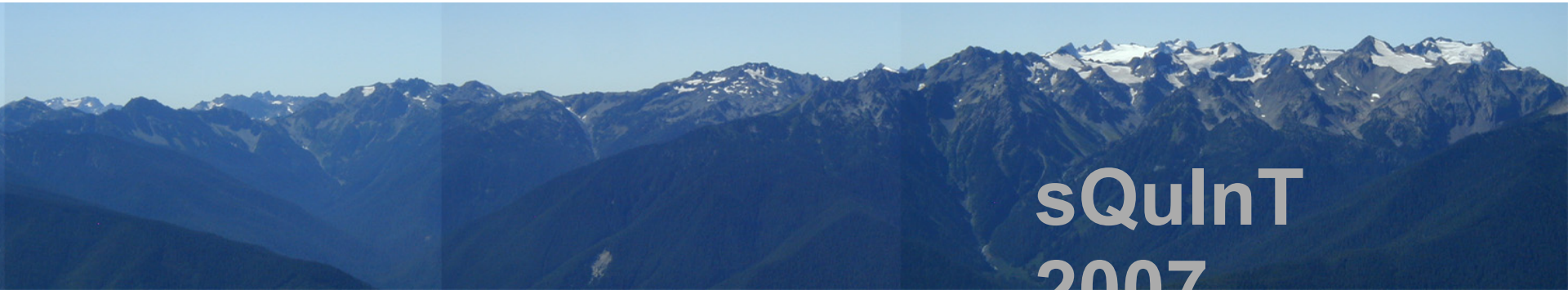


X error at location A causes increase of energy of Δ_2!

Do FT AQC's with forced syndromes lead to "natural" fault tolerance?

Is Fault-Tolerant Adiabatic Quantum Computation Possible?

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|QW⟩ group

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SCIENCE TO SHAPE THE FUTURE OF THE ARMY
ARMY RESEARCH OFFICE
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My Early Years





The Later Years Λ